RETAIL BUSINESS INTELLIGENCE

Executive Decision Report

Turning Online Retail II transactions into practical commercial decisions

Dataset: UCI Online Retail II | **Records:** approximately 1.07 million transaction rows

Analysis period: December 2009 – December 2011 | **Currency:** GBP (£)

Prepared as an executive-facing capstone deliverable

1. Executive Summary

A decision brief for leaders evaluating sales performance, customer value and commercial priorities.

Decision focus

Use transaction-level evidence to prioritize products, markets and repeat-customer activity.

What the asset provides

A large historical record of invoices, products, quantities, unit prices, customers and countries.

Recommended posture

Start with targeted pilots and measurable controls; do not commit to unverified revenue or profit claims.

Executive takeaway: The dataset is suitable for understanding historical sales mix, order patterns and customer purchasing behaviour. It does not contain marketing expenditure, product costs or inventory balances, so customer acquisition cost, true profitability and stock availability cannot be established from these records alone.

DECISION REQUESTED

Approve a 90-day pilot covering product/market prioritization, repeat-customer engagement and monthly performance monitoring.

2. Business Problem & Decision Questions

The project converts a transaction archive into a repeatable management information workflow.

Business question	Why it matters	Decision enabled
Which products contribute most to sales?	Revenue concentration can create dependency and opportunity.	Prioritize assortment, promotion and product availability.
Which countries generate demand?	Geographic performance can vary significantly.	Focus market development and service attention.
How do sales change over time?	Seasonality and peaks affect planning.	Schedule campaigns and operational capacity.
Who are the repeat and high-value customers?	Repeat purchasing supports more efficient engagement.	Build customer retention and reactivation tests.

Management objective: move from descriptive reporting to a small set of monitored actions with named owners, measurable baselines and review dates.

3. Dataset & Analytical Scope

Source: UCI Machine Learning Repository — Online Retail II.

Scale

Approximately 1,067,371 transaction rows, covering a two-year historical period.

Core fields

Invoice, stock code, description, quantity, invoice date, unit price, customer ID and country.

Unit of analysis

Invoice lines are not the same as unique orders or unique customers; metrics must be aggregated carefully.

Source link: UCI Online Retail II dataset. The data describes a UK-based online retailer and is historical, not a live operational feed.

Scope boundary: results describe recorded transactions only. They should not be interpreted as current market conditions or a complete view of company economics.

4. Methodology & Data Controls

A reliable executive conclusion depends on consistent treatment of transaction records.

Stage	Work performed / required	Control applied
Load & profile	Inspect columns, types, missingness, row counts and date coverage.	Retain a raw copy and record profiling results.
Clean	Standardize dates/text; inspect missing customer IDs and descriptions; identify duplicate rows.	Document exclusions and avoid silently dropping valid records.
Validate	Review cancellations/returns, negative quantities and non-positive prices.	Separate sales from cancellations/returns where appropriate.
Engineer	Create revenue value, month, year, invoice/order and customer-level measures.	Define each KPI once and reuse consistently.
Analyze	Trend, product, geography and customer analysis; visual checks and hypothesis tests.	Show denominator, period and filters with each result.

QUALITY PRINCIPLE
Do not treat every invoice line as a completed sale: cancellations, returns and unusual values must be checked before revenue is reported.

5. KPI Framework

Use a concise KPI dictionary so leaders can compare like with like.

KPI	Recommended definition	Interpretation / caution
Net sales value	Sum of quantity × unit price for validated sales lines, after defined return/cancellation treatment.	State currency and treatment of negative quantities.
Orders	Count of distinct valid invoice IDs.	Exclude cancelled invoices using a documented rule.
Average order value	Net sales value ÷ valid orders.	Sensitive to large orders and returns.
Units sold	Sum of valid sales quantities.	Separate returned units from sold units.
Active identified customers	Count of distinct non-missing customer IDs with valid purchases.	Customer ID is missing for some records; this is not total visitor count.
Repeat customer rate	Identified customers with 2+ valid orders ÷ identified purchasing customers.	A historical repeat measure, not a forward-looking churn metric.

6. Revenue Drivers to Examine

The dashboard is structured to isolate the main sources of sales contribution.

Product contribution

Rank products by net sales value and units. Compare the top contributors with the long tail.

Geographic contribution

Compare country-level sales and order counts; distinguish the home market from smaller export markets.

Time pattern

Review monthly sales and order movement to identify recurring peaks, troughs and unusual periods.

How to present the actual finding: show the leading products/countries/months with their value, share of total and comparison period. Avoid ranking only by quantity when the decision concerns revenue.

Current evidence status: the dashboard includes these views, but exact ranked values are not reproduced in this report because the validated notebook outputs were not available at report generation.

7. Customer Behaviour & Retention Opportunity

Customer IDs allow historical purchase-frequency and value analysis for the identifiable portion of the dataset.

Analysis	What to calculate	Possible management use	Caveat
Repeat purchase	Orders per customer and share with 2+ orders.	Test targeted reminders and loyalty communications.	Missing IDs reduce coverage.
Customer value	Historical net sales per identified customer.	Prioritize service and engagement for high-value groups.	Historical value is not future lifetime value.
Recency	Days since last valid purchase at dataset end.	Create a reactivation test cohort.	Period ends in 2011; not a current customer list.

RECOMMENDATION
Pilot customer reactivation using a small, measurable cohort; compare response and repeat orders against a holdout group.

8. Business Risks & Evidence Limitations

The report separates what can be measured from what requires additional business data.

Risk / gap	Why it matters	Mitigation / next data needed
Historical period only	Patterns from 2009–2011 may not represent current demand.	Refresh with recent transaction data before operational rollout.
No marketing spend	CAC and campaign ROI cannot be calculated.	Connect campaign costs, channel attribution and customer acquisition records.
No product cost / margin	High revenue does not necessarily mean high profit.	Add COGS, fulfilment, shipping and discount data.
No stock/inventory data	Cannot assess stockouts, overstock or inventory turns.	Integrate stock on hand, replenishment and purchase orders.
Missing customer IDs / cancellations	Can distort customer and sales measures if untreated.	Publish data-quality rules and KPI coverage rates.

Executive implication: use this analysis to prioritize questions and pilots—not as a substitute for current financial, marketing or inventory reporting.

9. Three Strategic Recommendations

Prioritize actions that can be tested with existing transaction fields and expanded with additional data.

1. Focus product and market effort

Use product and country contribution rankings to select a short list for merchandising and market-development tests. Track net sales, units and order count.

2. Grow repeat purchasing

Build a reactivation/retention pilot for identifiable past customers. Use a holdout group and measure repeat orders and incremental net sales.

3. Establish a trusted KPI cadence

Adopt a monthly dashboard review with fixed definitions, data-quality checks and an owner for each action. Add cost and inventory data next.

Decision rule: scale only when the pilot demonstrates measurable incremental benefit and the required data quality is acceptable.

10. Illustrative ROI Model — Not a Measured Forecast

A scenario framework for leadership discussion. Replace assumptions with validated baseline values before approval.

Because the dataset does not contain campaign spend, product cost or contribution margin, an actual ROI cannot be calculated. The following is a transparent planning example only.

Scenario input	Illustrative assumption	How to replace it
Eligible annual sales base	£1,000,000 (example only)	Insert validated recent annual net sales.
Incremental uplift from pilot	1.0%–2.0% (test, not promise)	Estimate from treatment vs holdout group.
Incremental sales	£10,000–£20,000	Base × measured incremental uplift.
Contribution margin	20% (example only)	Finance to provide product-level contribution margin.
Incremental contribution	£2,000–£4,000	Incremental sales × contribution margin.
Pilot cost	£1,500 (example only)	Include campaign, staff, tooling and fulfilment costs.
Illustrative net benefit	£500–£2,500	Contribution – pilot cost; validate before investment.

ROI FORMULA
ROI = (incremental contribution – pilot cost) ÷ pilot cost. Do not substitute revenue uplift for profit or contribution.

11. Implementation Timeline

A practical 90-day pilot plan with decision gates.

Phase	Timing	Key activities	Deliverable / gate
1. Validate	Weeks 1–2	Confirm current data, KPI definitions, cancellation/return rules and baseline.	Signed-off baseline and data-quality note.
2. Select pilot	Weeks 3–4	Choose product/market test and customer cohort; define holdout and costs.	Pilot brief, owners and success metrics.
3. Execute	Weeks 5–8	Run targeted activity; monitor weekly sales, orders, response and cost.	Interim dashboard and issue log.
4. Evaluate	Weeks 9–10	Compare pilot vs holdout; calculate incremental contribution and ROI.	Evidence-based scale / stop decision.
5. Embed	Weeks 11–12	Set monthly review cadence and backlog for cost/inventory integration.	KPI pack, governance owner and next-quarter plan.

Suggested owners: Commercial lead (pilot), Data analyst (measurement), Finance (margin/cost validation), Marketing/CRM (customer activity), Operations (inventory data integration).

12. Dashboard & Ongoing Governance

The Streamlit dashboard is the operational interface; the executive PDF is the decision record.

Dashboard

Interactive filters and views for revenue, average order value, monthly trend, country sales, top products and customer analysis.

Analytical notebook

Reproducible data cleaning, profiling, KPI calculation and exploratory analysis. Keep code, definitions and outputs aligned.

Monthly review

Review KPI movement, exceptions, data coverage, pilot outcomes and action owners. Record decisions and next steps.

Suggested dashboard controls: show selected date range, currency, data refresh date, treatment of cancellations/returns and number of identified customers on every relevant view.

Deployment: Streamlit Community Cloud deployment is the next publication step. Add the final public app URL to the portfolio once deployment succeeds.

13. Executive Decisions & Next Steps

The analysis provides a foundation for disciplined, evidence-led retail decisions.

Approve these three actions:

- Validate the KPI baseline using a documented treatment of cancellations, returns and missing customer identifiers.
- Run a controlled product/market and repeat-customer pilot over 90 days.
- Integrate marketing cost, product margin and inventory data before making CAC, profitability or stock-optimization claims.

SUCCESS MEASURES
Incremental net sales, incremental contribution, repeat-order rate, pilot ROI, data-quality coverage and on-time completion of milestones.

Portfolio evidence to submit

1. Executive Decision Report PDF (this document)
2. Final analytical Jupyter Notebook
3. Public Streamlit dashboard link
4. Supporting KPI dictionary and data-quality documentation

Repository: GitHub — Data-Analyst-Intern-Rab-Tech-

14. Sources, Assumptions & Reporting Notes

Read this page before using the report for investment or operational decisions.

Primary dataset

Chen, D. (2012). Online Retail II. UCI Machine Learning Repository. https://doi.org/10.24432/C5CG6D

Project repository

https://github.com/mahammadsajid/Data-Analyst-Intern-Rab-Tech-

Reporting notes

• Dataset scale and period are based on the project information supplied for this analysis.

• No precise product, country, monthly revenue or customer ranking is stated here because validated numeric outputs were not available in the source material reviewed for this report.

• The ROI example is illustrative and must not be represented as achieved or forecast company performance.

• The dataset is historical and should be supplemented with current business data before decisions are operationalized.

• CAC, actual churn, gross margin, profit, inventory turns and stockout rates require additional data sources.

USE LIMITATION
This is an analytical capstone decision framework, not audited financial reporting or a current commercial forecast.